In [2]:
```
#1.Sum of Even and Odd Numbers Get a number n from the user
# and calculate the sum of all even numbers and
# the sum of all odd numbers between 1 and n.
num=int(input("enter the number"))
es=0
ods=0
i=1
while(i<=20):
    if(i%2==0):
        es=es+i

    else:
        ods=ods+i
    i+=1
print(f"The even sum is:{es}\nThe odd sum is:{ods}")
```

```
The even sum is:110
The odd sum is:100
```

In [3]:
```
#2.Count Digits in a Number Get an integer from the user and count the number of
#Do not convert the number into a string. Example: 58392 → 5 digits
n=int(input("enter the numbers"))
count=0
num=abs(n)
if num==0:
    count=1
else:
    while num>0:
        num=num//10
        count+=1
print(count,"digits")
```

```
5 digits
```

In [6]:
```
3.#Find the Second-Largest Number Given a list of distinct numbers, find the sec
l1=[1,22,3,41,5,64,7]
l=l1[0]
sl=l1[0]
if sl>l:
    l,sl=sl,l
i=2
while i<len(l1):
    if l1[i]>l:
        sl=l
        l=l1[i]
    elif l1[i]>sl:
        sl=l1[i]
    i+=1
print("largest number:",l)
print("second largest number is:",sl)
```

```
largest number: 64
second largest number is: 41
```

In [17]:
```
#4.Count Vowels and Consonants Get a string from the user and count the vowels a
#Ignore spaces, digits, and special characters.
n=input("enter the any string:")
v=0
c=0
```

```
i=0
while i<len(n):
    ch=n[i].lower()
    if ch.isalpha():
        if ch in "aeiou":
            v+=1
        else:
            c+=1
    i+=1
print("Number of vowels:",v)
print("Number od consonants:",c)
```

```
Number of vowels: 3
Number od consonants: 4
```

In [18]:
```
#5.Calculate Factorial Get a positive integer from the user and calculate its fa
n = int(input("Enter a positive number: "))
fact = 1
i = 1
while i <= n:
    fact = fact * i
    i += 1
print("Factorial:", fact)
```

```
Factorial: 120
```

In [19]:
```
#6.Find Numbers Divisible by 7 but Not by 5 Print all numbers between 1 and 200
#Also, count the matching numbers.
i = 1
count = 0
print("Numbers divisible by 7 but not by 5:")
while i <= 200:
    if i % 7 == 0 and i % 5 != 0:
        print(i, end=" ")
        count += 1
    i += 1
print("\nTotal matching numbers:", count)
```

```
Numbers divisible by 7 but not by 5:
7 14 21 28 42 49 56 63 77 84 91 98 112 119 126 133 147 154 161 168 182 189 196
Total matching numbers: 23
```

In [20]:
```
#7. Calculate Electricity Bill Get the units consumed by each of five customers
# * First 100 units: ₹2 per unit
# * Next 100 units: ₹3 per unit
# * Above 200 units: ₹5 per unit
units = []
i = 0
while i < 5:
    u = int(input("Enter units consumed: "))
    units.append(u)
    i += 1
i = 0
while i < 5:
    u = units[i]
    if u <= 100:
        bill = u * 2
    elif u <= 200:
        bill = (100 * 2) + ((u - 100) * 3)
    else:
        bill = (100 * 2) + (100 * 3) + ((u - 200) * 5)
```

```
    print("Customer", i + 1, "Bill: ₹", bill)
    i += 1
```

```
Customer 1 Bill: ₹ 160
Customer 2 Bill: ₹ 260
Customer 3 Bill: ₹ 440
Customer 4 Bill: ₹ 800
Customer 5 Bill: ₹ 1000
```

In [21]:
```
#8.Reverse a Number Get an integer from the user and reverse its digits using a
#Do not convert it into a string. Example: 12345 → 54321
n = int(input("Enter a number: "))
num = abs(n)
reverse = 0
while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10
if n < 0:
    reverse = -reverse
print("Reversed number:", reverse)
```

```
Reversed number: 54321
```

In [22]:
```
#9.Analyze Student Marks Given a list of marks, count the students who passed (5
marks = [85, 45, 90, 35, 70, 55, 40, 95]

passed = 0
failed = 0
distinction = 0
i = 0
while i < len(marks):
    if marks[i] >= 50:
        passed += 1
    else:
        failed += 1
    if marks[i] >= 75:
        distinction += 1
    i += 1
print("Students Passed:", passed)
print("Students Failed:", failed)
print("Students with Distinction:", distinction)
```

```
Students Passed: 5
Students Failed: 3
Students with Distinction: 3
```

In [23]:
```
#10.Find Perfect Numbers Print all perfect numbers between 1 and 1000 using a si
#A perfect number equals the sum of its proper divisors. Example: 6 is perfect b
print("Perfect numbers between 1 and 1000:")
while num <= 1000:
    i = 1
    total = 0
    while i < num:
        if num % i == 0:
            total += i
        i += 1
    if total == num:
        print(num)
    num += 1
```

```
Perfect numbers between 1 and 1000:
0
6
28
496
```

In [ ]: